KENA TUITOEK

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EDUCATION

University College London *London, UK*
MSc Data Science and Public Policy (Economics track) 2025 – 2026

- Modules: Time Series Econometrics, Statistical Learning for Programming, Machine Learning in Economics, Quantitative Text Analysis, Data and Evidence for Public Policy

University of Sussex *Falmer, UK*
BSc Economics, First-Class Honours, Award: Highest Applied Economics Dissertation 2022 – 2025

- Modules: Applied Econometrics, Advanced Microeconomics, Industrial Organisation, Economics & Data Science, Data Coding & Visualisation

RESEARCH PROJECTS

Corporate Communication and Systemic Risk Prediction *Python*

- Fine-tuning FinBERT on earnings call transcripts from the 10 largest US financial institutions (2010–2024) to classify management tone across four dimensions (sentiment, uncertainty, risk language, forward-looking negativity), aggregating into an SRISK-weighted firm-level stress sentiment measure.
- Testing whether the SRISK-weighted sentiment measure predicts next-quarter systemic risk (SRISK, ΔCoVaR) beyond standard market controls using panel methods and out-of-sample forecasting, a form of early warning signal.
- Built automated pipeline for SEC EDGAR filings and earnings call transcript extraction; processed 5,000+ documents with speaker-turn segmentation for management vs. analyst sections.

WTI Crude Oil Price Forecasting with Geopolitical Risk Augmentation *Python*

- Developed a stacked ensemble (gradient boosted regression, random forest) for monthly WTI price direction forecasting, augmented with the Caldara-Iacoviello Geopolitical Risk index and OVX implied volatility.
- Built a full backtesting engine with CME contract specifications, half-Kelly position sizing, and walk-forward out-of-sample validation. Results: 11.2% total return, 0.353 Sharpe ratio; high-volatility trades earned 4.6× more per trade.
- Conducted GPR-augmented model comparison using Diebold-Mariano testing; built an interactive Chart.js dashboard for real-time signal monitoring and performance attribution.

Carbon Pricing and Trade Competitiveness in Steel Sectors *R, Stata*

- Merged UN Comtrade bilateral trade flows (HS chapters 72–73) with World Bank carbon pricing data; handled ISO3 harmonisation and subnational scheme aggregation (EU ETS, RGGI).
- Estimated carbon price–export elasticities using two-way fixed effects panel regression on 216 country-product-year observations. Complemented with PPML gravity specification to address zero trade flows.
- Results support the implementation of CBAM border adjustments and free allowances to mitigate carbon leakage in energy-intensive trade-exposed (EITE) steel sectors.

Gravity Model of Africa's Bilateral Trade *Stata, R*

- Estimated PPML gravity model with importer-year, exporter-year, and pair fixed effects on 960K pair-year observations (CEPII, 1960–2021); stress-tested specifications with border-year controls and 2/4/6-year lag structures.
- Built an end-to-end reproducible data pipeline with joint WTO and disaggregated PTA indicators.
- Found WTO membership increases African bilateral imports by 15–19%, customs unions deliver 20–25% persistent gains, while shallow FTAs show negligible impact; findings inform regional trade integration recommendations.

PROFESSIONAL EXPERIENCE

UKRI Data Science DRI — Data Visualisation Training Workshop *2025*

- Built an interactive data visualisation suite (Plotly, Bokeh, Altair, Streamlit) for multi-stakeholder audiences. Implemented modular data loaders and reusable chart components for rapid data integration.
- Applied ML pipelines with custom visualisations for model diagnostics (calibration curves, feature importance, uncertainty quantification) for non-technical decision-makers.

Nemea Fund — Analyst *Oct 2023 – Jun 2025*
Student-managed investment fund, University of Sussex Falmer, UK

- Extracted and synthesised macroeconomic and industry data from Bloomberg Terminal to support quantitative investment recommendations. Contributed to sector allocation decisions and portfolio risk reporting.
- Conducted consumer discretionary sector analysis, equity research and evaluated competitive positioning, earnings trends, and macro sensitivity to inform portfolio allocation.

TECHNICAL SKILLS

- **Languages:** Python (pandas, scikit-learn, statsmodels, PyTorch, Hugging Face), R (tidyverse, fixest), Stata, SQL
- **Methods:** Panel data (FE, PPML), time series (SVAR, Bayesian VAR, Kalman filter), local projections, causal inference (DID, synthetic control, DML), ML (XGBoost, LightGBM, CNN, BiLSTM, Transformers), NLP (FinBERT, TF-IDF, topic models)
- **Tools:** Bloomberg Terminal, WRDS, IMF/World Bank APIs, UN Comtrade, LOBSTER, Git, LaTeX, Streamlit